Open Concerns

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26 October 2026

Every passage the final report marks as queried, and the status of each of its eight result slots, collected into one list. 98 items, from the final report.

A flag marks a result, number or condition that is *measured but not yet trustworthy as evidence* — an unexplained mechanism, a confound, an unfair comparison, or a figure whose provenance does not support the use it is being put to. It deliberately does not mark a bad result: a large error that is correctly measured and understood is a finding, not a concern.

The report's source still carries every one of these passages beside the claim it qualifies, but the report no longer renders them: queries are typeset as plain text or dropped, and the status boxes are dropped. This document is where they are read. References such as `[sec:example]` point into the report.

Front matter

1. [To be written by the author. The template reserves this page for thanks to the project advisor, the examination committee, Gensurv Robotics, the Institute of Field Robotics, and any funding source. Nothing has been drafted here on the author's behalf.]

Chapter 3 — Methodology

The Pipelines as Implemented: ORB-SLAM3 and semantic feature rejection

1. *table row:* `loopClosing` | 1 | comment in the same file says "set to 0 deliberately"

Simulation Platform: Computing and software environment

2. *table row:* OpenCV (Python) | 4.11.0 | differs from the 4.5.4 the estimators link

Simulation Platform: Simulated sensor rig and platform

3. *table row:* `acc_n` | 0.02 | 0.0139–0.0294 | $\mathrm{m\,s^{-2}}/\sqrt{\mathrm{Hz}}$

4. *table row:* `gyr_n` | 0.002 | 0.000300–0.000364 | $\mathrm{rad\,s^{-1}}/\sqrt{\mathrm{Hz}}$

5. [Queried] The configured gyroscope noise density is 2×10^{-3} whereas the stationary windows of the five recorded streams give 3.00×10^{-4}–3.64×10^{-4}, so the configured value is about six times too large. The configured accelerometer density of 0.02 sits inside the measured spread but that spread is itself wide, 0.0139–0.0294 across five bags of the same simulated sensor. Both estimators are therefore running on an inertial noise model that does not describe this IMU closely. The bias random walks cannot be checked at all: the longest stationary interval in any `dataset_allsensor` bag is 1.54 s and an Allan-variance estimate needs minutes. No conclusion in this report depends on these four numbers, but they should be re-derived from a dedicated stationary recording before any tuning claim is made.

6. [Queried] The accelerometer characterisation is not stable across recordings of the same simulated sensor. `allsensor_002`, whose stationary window is 1.54 s and therefore the best conditioned of the three, returns an x bias of $0.0683\,\mathrm{m\,s^{-2}}$, against 0.0191 for `allsensor_000` and $0.0016\,\mathrm{m\,s^{-2}}$ for `allsensor_004`, whose window is the shortest at 0.52 s. The scale ranges from 1.0199 to 1.0669 over the five. Because the window is short in every case, part of this spread is estimator variance rather than a property of the sensor, and the accelerometer terms should not be quoted as calibrated constants. The gyroscope scale agrees to four decimal places across all five bags; its bias does not, varying by a factor of twenty, which is consistent with the same window-length problem.

Simulation Platform: Simulation fidelity and its limits

7. *table row:* Accelerometer, per axis ($\mathrm{m\,s^{-2}}$) | 0.0566 | 4.002×10^{-3} | 1.39×10^{-2}–2.94×10^{-2} | 2.00×10^{-3}

8. *table row:* `slip1`, `slip2` | 0 on all grounds | an ODE feature; `dartsim` does not implement it, so inert at any value

9. [Threat to validity] This is a bias with a known direction. The EKF baseline of [sec:results-proprio] is evaluated on a floor where the wheels effectively cannot slip and with an encoder of unlimited resolution, which is dead reckoning's best case. Both idealisations favour the EKF over the visual estimators, and neither has been relaxed. The conclusion that a wheel+IMU filter outperforms both visual pipelines should be read as holding *under these conditions*, not as a general result. `script/make_slippery_floor.py` generates ground models at a chosen friction coefficient so that traction can be made an experimental factor; `script/encoder_sim.py` imposes a realistic quantisation floor. Neither has been used in any result reported here.

10. *table row:* `gyr_w` | 0.0001 | 1.735×10^{-6} | 58× | gyroscope bias random walk

Simulation Platform: Simulation environments and controlled factors

11. [Naming] The factor recorded in this project as "textured versus featureless floor" is narrower than the name suggests, and [fig:floor-treatments] is the evidence. Both conditions rest on the *same flat,*

untextured diffuse colour; the stock concrete-grain texture appears in neither. The entire difference is fourteen painted bay outlines, 0.145 m wide, covering a small fraction of a 1704 m^2 floor. A robot driving an aisle sees a uniform grey surface in both conditions for most of its path, crossing a high-contrast strip only at a bay boundary.

Results comparing the two floors must therefore be read as *bay markings versus nothing*, not as *realistic floor versus nothing*, and the near-absence of a visual-odometry improvement between them ([sec:results-proprio]) should not be generalised to floors carrying genuine surface texture. Testing that would require a world built on the stock `GroundB_01` model, which no recorded dataset uses.

12. [Unverified] The dynamic world's baked grid, [fig:warehouse-maps](c), contains 38 % fewer occupied cells than the distributed static arrangement in (a). `bake_map.py` rasterises every `<include>` in the world file at its authored pose, so kinematically driven props should appear at their initial positions, but this has not been confirmed for the dynamic family. Until it is, panel (c) must not be used as occupancy truth for scoring a reconstructed map of a dynamic scene: the deficit may be a genuine difference in static stocking, or it may be props missing from the bake, and those two possibilities imply opposite corrections. The estimator results in [sec:results-proprio] are unaffected, as they use the arrangements in panels (a) and (b) only.

13. *

14. *

15. *

16. [Missing world] `dataset_static_nofloortexture_000` and `_002` name `worlds/small_warehouse_static/small_warehouse_static_nofloortexture.world` in their run manifests, and that file no longer exists in the repository. Both datasets contribute to the relogged campaign of [sec:results-validity], so their scenes could not be drawn from the recorded world and their conditions cannot be re-inspected directly.

The panels marked * are reconstructions. They are exact for geometry rather than approximate, because `script/make_plain_floor.py` produces a floor variant by swapping only the ground model's `<include>` URI and leaves every object untouched: the missing world therefore differed from the surviving `small_warehouse_static.world` in its floor alone, and that floor is recoverable from the suffix. It remains a reconstruction and not the recorded file, and the two datasets should be re-recorded, or the world restored from history, before either is used for anything the floor or the object layout bears on.

Simulation Platform: Realistic logistics layouts

17. [Visual verification] The generator-side checks are complete, and the first Gazebo session on 6 October 2026 found and fixed one defect the checks could not see (a jack setting its load down and then stopping on the neighbouring cell, because the drop command carried one angle for both the parked load and the approach heading). Three things remain to be confirmed by eye in the simulator rather than by the checker: that the loads ride on the fork side of the jack, the shading of the generated door, trailer and yard meshes, and the open/closed state of each door. Until then the worlds are **implemented** in geometry and **partial** in visual verification.

18. *table row:* Run manifest | none

19. [Provenance] `dataset_map_realistic` was recorded without a run manifest, so its world is not recorded anywhere. It was identified from the bag itself: the `/world/default/pose/info` stream names 129 models whose numbering and composition match the regenerated U-shape world and no other, and every model except the robot holds the same pose from the first message to the last. That is consistent with the static twin, but also with the dynamic world loaded without its traffic plugin, and the bag cannot tell

the two apart. The exact `.world` file must be confirmed by the operator and a manifest written before the recording is used in any result slot; the identification above is evidence, not provenance.

Simulation Platform: Dataset registry

20. *table row:* `dataset_static_nofloortexture_001` | static | 617 | 1 | 92.0 | 113 332 | ✓ | ✓ | – | FD stub, WS ok

21. *table row:* `dataset_dynamic_nofloortexture_01_000` | dynamic `_01` | 713 | 710 | 165.4 | 126 790 | ✓ | ✓ | – | FD malformed, WS ok

22. *table row:* `dataset_real_001` | real rig | – | 2099 | 85.5 | 9 629 | – | – | – | 0 IMU messages

23. *table row:* `dataset_real_002` | real rig | – | 2869 | 113.5 | 12 682 | – | – | – | 0 IMU messages

24. [Data integrity] Two drive copies are unreadable while presenting a complete, plausible `metadata.yaml`. `dataset_static_nofloortexture_001` is a 524 288 B stub whose metadata nevertheless declares 113 332 messages, and `dataset_dynamic_nofloortexture_01_000` is 710 MB—within 0.4 % of its workspace counterpart—with a valid SQLite header but a store that reports *database disk image is malformed* on any query. Both have intact workspace copies, which is why those two datasets were read from the workspace in the relogged campaign.

The general lesson is recorded as a control rather than an incident: `metadata.yaml` is written when recording begins and does not describe what survived, so message counts, durations and topic lists copied from it are not evidence that a bag is intact. Storage health must be established by opening the SQLite store and counting rows, which is now part of the pre-experiment inspection in [sec:collection-validity].

25. [Scope limit] Only the five `dataset_allsensor` bags carry wheel encoders and wheel odometry; the other thirteen recordings omit those topics entirely. The proprioceptive baseline of [sec:results-proprio] therefore cannot be computed for any dynamic condition, for the mapping bags, or for the real rig, and the comparison between proprioception and visual odometry exists only in *static, featureless-floor* scenes—the conditions least favourable to vision and most favourable to dead reckoning. Extending it requires re-recording with the two topics bridged, not new analysis.

26. [Excluded] `dataset_real_001` and `dataset_real_002` are excluded because their inertial topic `/hwt101ct_yaw_publisher` contains *zero* messages, so neither supports stereo-inertial operation at all. Their image streams are intact (2 562 and 3 405 pairs), so they remain usable for detector or image-only work. `dataset_real_000` is the only real recording with inertial data, at 23 257 messages.

Real Platform: AMR mobile robot testbed

27. The drivetrain is a rear brushless motor (Flipsky 6384) through a BLDC drive with a Dynamixel RX-64 steering servo, a GigaDevice GD32407V microcontroller on a CAN link to the NRU-51V, an LM2596 regulator from a lithium-polymer pack, and a Wi-Fi router for tele-operation from a smartphone. The faculty template describes the base as a differential-drive robot; the hardware diagram (a single traction motor with a steering servo) and the simulated model of **??** are Ackermann. The description here follows the hardware and should be confirmed by the author.

Real Platform: Real sensor rig and calibration limits

28. *table row:* Second-camera intrinsics | reuse of the first camera | approximation

29. *table row:* IMU-to-camera translation | zero | placeholder

30. *table row:* Stationary $\|a\|$ | $9.2642\,\mathrm{m\,s^{-2}}$ | measured, deviates from 9.81

The Four Experiments: Experiment 1: Non-visual baseline and estimator comparison

31. *table row:* accelerometer x bias | 0.0191 | 0.0304 | 0.0683 | 0.0309 | 0.0016 | $\mathrm{m\,s^{-2}}$

32. *table row:* accelerometer x scale | 1.0199 | 1.0249 | 1.0669 | 1.0525 | 1.0465 | –

33. *table row:* accelerometer x scale | 1.0199 | 1.0249 | 1.0669 | 1.0525 | 1.0465 | –

34. *table row:* accelerometer x scale | 1.0199 | 1.0249 | 1.0669 | 1.0525 | 1.0465 | –

35. *table row:* yaw-rate scale, bicycle | 1.0110 | 0.9527 | 1.0034 | 0.9375 | 0.9250 | –

36. *table row:* yaw-rate scale, bicycle | 1.0110 | 0.9527 | 1.0034 | 0.9375 | 0.9250 | –

37. *table row:* Window | static duration | 0.68 | 0.58 | 1.54 | 1.04 | 0.52 | s

38. σ_b, the bias random walk, is assumed rather than measured: the longest stationary window in any recording is 1.54 s and an Allan-variance estimate needs minutes.

Planned Versus Implemented

39. *table row:* Real-robot evaluation | metric accuracy against ground truth | calibration-limited, no independent trajectory truth | blocked

40. *table row:* Metric infrastructure | resource, throughput, uncertainty, manifests | partially implemented; RPE now complete | incomplete

Chapter 4 — Experiments and Results

Experiment 1: Non-Visual Baseline and Estimator Comparison: Simulation

1. **Status: completed.** Five `dataset_allsensor` recordings spanning four worlds and both floor conditions; three proprioceptive estimators, one transferred-noise EKF, VINS-Fusion, and ORB-SLAM3 in two loop-closure configurations, thirty-five runs, all passing their predefined checks.

2. *table row:* IMU noise model | not controlled in this experiment. The EKF's is measured per bag by `proprio_estimator.py calibrate`; both visual estimators read four constants from `config/wil_sim/stereo_imu.yaml`. The factor is confounded with the estimator here and is separated in [sec:proprio-tuning]

3. The initialization and reset figures below were extracted for `allsensor_000` and `allsensor_001` only and have not been recovered for the other three bags: VINS-Fusion initialized once in 0.33 s with no resets, and both ORB-SLAM3 runs shut down gracefully with no tracking-loss events and inertial initialization at 2.48 s and 2.51 s.

4. [Queried] ORB-SLAM3 receives fewer stereo pairs than are available on every dataset, while VINS-Fusion receives all of them (2639, 2478 of 2479, 2628, 3102 and 3046). The ORB shortfall is 3.8–11.9 % and varies by run as well as by dataset: 90.2 % and 94.2 % of the available pairs on `allsensor_000` for the two configurations, 96.2 % and 95.1 % on `allsensor_003`. Its own counters report zero decode failures, and the stereo timestamps are aligned to 0 s median offset with 100 % of pairs inside the 20 ms synchronisation slop, so synchronisation is not what loses them.

The counter itself cannot say where they go: `stereo_pairs_received` is incremented inside the *synchronised* callback, and the node keeps no count of images arriving. What the source does show is an asymmetry in input buffering. ORB-SLAM3 subscribes with `rclcpp::SensorDataQoS()`, which is best-effort with a depth of five, so a frame arriving while the tracking callback is busy is discarded by the

middleware and never recorded anywhere; VINS-Fusion subscribes with `KeepLast(100).reliable()` and buffers the same burst. The shortfall is therefore most consistent with ORB-SLAM3 dropping frames at its own subscriber under load, not with loss in transport. The loop-closure-free runs support that reading on three datasets, receiving more pairs than the enabled runs on `allsensor_000`, `_001` and `_002` while doing less work per frame, though `allsensor_003` runs the other way and the mechanism is not proven. Trajectory accuracy over a shared interval is not obviously affected—ORB-SLAM3 is the more accurate of the two on some datasets despite the shortfall—but no throughput, latency or per-frame compute comparison between the two estimators is admissible until the loss is explained: the two estimators differ not only in load but in input-queue policy, so such a comparison would measure the QoS choice as much as the algorithm.

5. [Ruled out] A plausible reading of these results is that the visual estimators lose to the EKF because their input rate collapsed. The data does not support it. Source imagery is regular at 30.3 Hz in simulated time on all five bags, with a median inter-frame interval of 0.0330 s and no gap exceeding 0.1 s; VINS-Fusion received and processed every frame on every dataset, with zero skipped before the back end, and still trails the cross-validated EKF on four of five. Frame loss is real for ORB-SLAM3 and remains unexplained, but it cannot account for the margin, because the estimator with no frame loss at all shows the same one.

6. [Queried] ORB-SLAM3 was run in two configurations on every dataset, recorded as `logging_20260916_01` with loop closing enabled and `logging_20260916_02` with it disabled, and both are reported below as separate systems. Within the enabled configuration the loop closer behaves very differently per dataset ([tab:orb-loop]), so those five trajectories are not repeats of one condition and must not be pooled.

A counter reading `loop_closures,0` does *not* mean the loop closer was inactive. On `allsensor_000` it performed 447 place-recognition checks, found 90 loop candidates and rejected all 90 on inertial geometry; on `allsensor_004` it found 114 and accepted none. The disabled runs are distinguishable only in `loop_closing.csv`, where every event carries `current_keyframe_id` $= -1$ and a duration below 0.003 ms, the signature of a thread spinning with nothing queued to it.

7. [Provenance gap] `run_metadata.csv` records 22 parameters and loop-closure state is not among them, so the whole distinction rests on the directory name—and that name does not mean the same thing everywhere in this chapter. In the relogged campaign of [sec:results-validity], `logging_20260916_01` through `_03` are three *repeats* of one configuration, all with an active loop closer. In the `allsensor` campaign reported here, `_01` and `_02` are two *configurations*, loop closing enabled and disabled, with one run each. The same suffix therefore encodes a repeat index in one section and an experimental factor in the next. The metadata for the two bag-000 runs is identical apart from the experiment identifier, output path and start time, and both name the same installed configuration file, so nothing in the recorded provenance distinguishes an enabled run from a disabled one. The distinction used in this chapter was recovered from `loop_closing.csv` after the fact. The logger should record the loop-closing and global-BA switches explicitly before any further ORB-SLAM3 campaign.

8. *table row:* plain | Wheel odometry | 1.485 | 2.888 | 4.332 | 3.76 | −30.11

9. *table row:* `allsensor_003` | `bayline_` | IMU dead reckoning | 45.562 | 62.903 | 74.745 | 53.16 | −5.84

10. *table row:* `objonwallonly` | Wheel odometry | 3.192 | 3.979 | 9.231 | 6.57 | −37.17

11. *table row:* ORB-SLAM3 | 6.481 | – | – | – | –

12. That earlier figure of "3.0, 1.2 and 2.2 times" was computed against ORB-SLAM3 with loop closing enabled, which is not the best visual result available. Measured against the best visual estimator of each dataset, and using the cross-validated EKF ([tab:proprio-xval]), the margins are 1.38, 1.41, 2.53, 1.31 and 1.13.

13. Nor did adding the bay markings help it —the two `bayline` datasets give VINS-Fusion 0.784 and 0.425 m against 0.616, 0.383 and 0.558 m on the plain floors, which is no improvement.

14. H1 is falsified across all five datasets.

15. *table row:* `allsensor_002` | +0.34 | −30.11 | 865 | −3.481 | −31.03

16. *table row:* `allsensor_003` | −6.25 | −37.17 | 1123 | −3.309 | −38.51

17. [Corrected] An earlier reading of this experiment proposed that the offset came from the bicycle model averaging the two kingpin angles, since tan of a mean is not the mean of tan. That explanation is *wrong* and is withdrawn. Recomputing the yaw rate as the mean of each wheel's own bicycle-equivalent rate changes the residual from −0.00277 to −0.00288, from −0.00285 to −0.00296, and from −0.00624 to −0.00630 rad s^{-1}— under 4 % in every case, two orders of magnitude short of what the observed drift requires. The averaging is not the source.

18. *table row:* `allsensor_003` | bayline | 0.9375 | 0.1192 | 3.54 | 0.423

19. *table row:* `allsensor_004` | bayline | 0.9250 | 0.0613 | 3.41 | 0.494

20. That inference does not survive the textured-floor datasets and is withdrawn: the coupling is the dominant term in the EKF's heading error whenever the wheel rate is poor, and it sets a floor on the filter's accuracy that no amount of gyroscope quality can lift.

21. *table row:* IMU dead reckoning | 311.0 | 0.005 | 0.022 | 0.043 | overconfident

22. *table row:* IMU dead reckoning | 339.9 | 0.001 | 0.009 | 0.021 | overconfident

23. *table row:* IMU dead reckoning | 244.7 | 0.017 | 0.045 | 0.147 | overconfident

24. *table row:* IMU dead reckoning | 1504.3 | 0.010 | 0.018 | 0.035 | overconfident

25. *table row:* IMU dead reckoning | 1504.3 | 0.010 | 0.018 | 0.035 | overconfident

26. *table row:* IMU dead reckoning | 289.5 | 0.004 | 0.017 | 0.044 | overconfident

27. [Method caveat] The χ^2 interval behind the verdict column assumes independent samples. At 200 Hz consecutive dead-reckoning errors are almost perfectly correlated, so the test is run on a 1 Hz subsample (81 to 102 samples per dataset). That reduces the dependence but does not remove it, and the verdicts should be read as indicative. The σ-coverage columns make no independence assumption and are the sturdier statement.

28. [What this control does not cover] Exchanging noise models between bags tests whether the filter saw *this* recording's answer. It cannot test the asymmetry that matters here, because both models are still measured from data: all five bags come from one simulated IMU, so the exchanged σ values differ by less than a factor of two on every term except accelerometer x. The visual estimators receive nothing of the kind—they read four constants from a file. The control in [sec:proprio-tuning] replaces this one.

29. [What the control does and does not establish] H4 was written down after the exploratory runs had been scored, not before. What is reported here is a post-hoc formalisation of an exploratory finding; the decision rule and the direction of the test were fixed before this subsection was written, but not before the data was seen. The unanimity—every bag, both directions, a mechanism that isolates to one term—is what makes it worth reporting at this stage, not a pre-registered test, and a confirmatory repeat on bags not used to form the hypothesis is the honest follow-up.

This is also an equal-handicap comparison, not an equal-effort one: both families now run on a declared IMU noise that nobody fitted to the data. It establishes that the EKF's advantage in [tab:proprio-accuracy] is smaller than the tuning difference between the two families, and reverses under it. It does *not* establish that

the visual systems would win a symmetric comparison. The mirror experiment—giving VINS-Fusion and ORB-SLAM3 the measured values—requires re-recording and has not been run to a reportable standard; the one VINS run whose metadata confirms it diverged ([sec:sim-fidelity]). The transfer is not perfectly symmetric in role either: the filter's σ_g is a scalar in a planar five-state model, whereas `gyr_n` feeds three-axis pre-integration. Finally, $n = 1$ per cell is a property of a deterministic filter and not a design choice, but it means the between-bag range is the only uncertainty this experiment can report.

30. [Generalisation limit] The comparison is confined to *static* scenes. Four worlds and both floor conditions are now covered and the ranking is stable across them, so the earlier form of this limitation—that the result held only on featureless floors, vision's worst case—is no longer the binding one. Adding the bay markings did not improve either visual estimator, which removes that explanation rather than confirming it—though the markings are a much weaker manipulation than "textured floor" suggests ([fig:floor-treatments]), so a genuinely textured surface remains untested.

What remains is that every dataset carrying wheel encoders is static ([tab:dataset-registry]). The result must not be restated as "proprioception outperforms visual SLAM" without that qualifier, and it cannot be extended to dynamic scenes—where wheel odometry is unaffected by moving objects, and where the visual systems' known failures occur—until the `allsensor` topics are recorded in a dynamic world. That recording remains the single most informative next run in this line of work.

31. [Queried] Both visual estimators replayed their bags at 0.5× real time, so each frame received roughly twice the wall-clock budget it would have online. The accuracy figures in [tab:proprio-accuracy] are therefore not evidence of real-time capability, and no timing conclusion may be drawn from this experiment. The three baselines, being offline, have no replay rate at all, which is a second asymmetry in the same comparison.

32. [Statistical limit] $n = 5$ across four worlds, and no condition is repeated more than twice. The tables therefore report individual values deliberately; no mean, standard deviation or confidence interval is computed, and between-dataset differences are not evidence of a systematic effect. A single run per world cannot establish a scene-content effect; it shows only that the estimator ranking survives changes of scene.

33. [Unmeasured variance] ORB-SLAM3 is multi-threaded and not deterministic, and its run-to-run spread on identical input has not been measured here: each of its two configurations was run once per dataset. Several differences this chapter reports sit within a range that spread could plausibly cover—most sharply 6.481 m with loop closing enabled on `allsensor_003` against 0.515 m with it disabled, and the reversal on `allsensor_002` where disabling it is three times *worse*. Those differences are real measurements of the runs that were made, but they cannot yet be attributed to the loop-closing switch. Three or more repeats per configuration, reported as median and spread in the manner of the relogged campaign ([sec:results-validity]), are required before any loop-closure conclusion is drawn.

Experiment 1: Non-Visual Baseline and Estimator Comparison: Real robot

34. **Status: partial.** One real recording with an independent planar ground truth, scored after the cutoff (20 October 2026). Visual estimators only: the real platform publishes no wheel odometry in metric units and no joint states, so the non-visual baselines of [sec:results-proprio] cannot be formed. Every value is provisional until the marker-to-camera lever arm is measured.

Experiment 2: Offline YOLO Detector Performance: Simulation

35. **Status: completed.** Four dynamic recordings, 9 860 left-camera frames, scored against projected simulator ground truth at the thresholds of [tab:yolo-eval-params].

Experiment 2: Offline YOLO Detector Performance: Real robot

36. **Status: planned.** Real imagery exists but no real ground-truth boxes do.

Experiment 3: Visual-Inertial Odometry With Dynamic-Environment Filtering: Simulation

37. **Status: partial.** Two testbeds. The extreme world (the scripted `small_warehouse_dynamic_*_nofloortexture` warehouses, September 2026 campaign, 66 runs) left the filtering hypothesis untested: the VINS-Fusion filter never activated, eleven of twelve ORB-SLAM3 treatment pairs were not input-matched, and the world itself was found not to be a valid testbed. The realistic world (physically simulated traffic, runs of 14–21 October 2026) gives the environment-dynamics factor its first valid test, from one run per condition; its filtering arms yield no admissible pair either.

Experiment 3: Visual-Inertial Odometry With Dynamic-Environment Filtering: Real robot

38. **Status: blocked.** Real-robot trajectory accuracy cannot be claimed at the cutoff: the rig's IMU-to-camera translation is a zero placeholder, the second camera reuses the first camera's intrinsics, and the one recording with an independent ground truth, added after the cutoff, has not been run under this experiment's protocol ([tab:real-rig,tab:deviations]).

Experiment 4: Dense Map Reconstruction and Map-Quality Evaluation: Simulation

39. **Status: partial.** Three databases are scored for geometry and drivability: the tiny warehouse, which passes, and two realistic-warehouse maps, one per layout, built on 26 October 2026 with loop closure deliberately disabled, which fail the quality criterion fixed before scoring. The tiny map's independent drivability reference remains lost.

40. **[Queried: reproducibility of the drivability half]** The geometry result is fully reproducible: both `map/map_tiny.db` and the world file are present. The drivability result is not. Its reference run, `dataset_static_tiny_repeat`, is absent from `dataset/` and from the removable drive, and the only surviving recording of this warehouse is the one that built the map, which would answer a weaker question ([sec:map-drivability]). The database additionally stores an empty parameter set, so the configuration that produced it is not recoverable from the file, and the RTAB-Map engineering record's claim that the node stamps match no dataset is contradicted by the pose agreement reported above. The drivability figures should therefore be read as a dated measurement rather than a standing result, and should not be carried into the major report until an independent route has been recorded and retained.

41. The failure was attributed there to the front end under repeated structure, not to the mapping back end. No database, run directory or figure for this attempt is registered in the repository at the cutoff, so the result is reported on the strength of the progress record and cannot yet be audited. The attempt must be repeated with its artifacts retained before it is carried into any conclusion.

42. **[Queried: front end of the realistic builds]** RTAB-Map took its pose from TF, so the odometry came from whichever front end was running; the database does not record which. The precision loss cannot be placed on the front end against the back end until the build logs name it.

Experiment 4: Dense Map Reconstruction and Map-Quality Evaluation: Real robot

43. **Status: planned.** No RTAB-Map database has been built from real imagery, and no reference geometry of the site exists to score one against.

Cross-Experiment Discussion, Simulation-to-Real Comparison and Limitations: Where each pipeline is weak, block by block

44. [Withdrawn criterion] An earlier revision of these tables graded VINS-Fusion's solver quality against "median final cost > 5000". That threshold was chosen after observing 1176 in static scenes and 14110 in dynamic ones, so it separated two numbers already in hand and the "poor" verdict it produced was guaranteed by the choice rather than tested by it. Raw Ceres cost also has no absolute scale, depending on residual count and weighting, so no fixed threshold on it is meaningful in principle. It is replaced by the solver's own `solver_solution_usable` flag, which is binary and emitted by the solver rather than invented here: 99.64 % of solves in static scenes and 99.55 % in dynamic ones return a usable solution, and the block is now graded good in both. The non-convergence rate—22.9 % of solves in static scenes and 45.3 % in dynamic ones reaching the iteration or 80 ms cap—is reported in the note column and deliberately not graded, because any threshold on it would repeat the same error.

45. Both visual front ends are the dominant cost, and neither keeps up cleanly. ORB-SLAM3 spends 33.1 ms of a 53.8 ms tracking budget on preprocessing, extraction and stereo matching in static scenes, rising to 45.1 ms of 61.2 ms in dynamic ones. VINS-Fusion's back end waits 403.7 ms per frame on its feature queue in static scenes and 502.1 ms in dynamic ones, against a 66.0 ms frame budget, with a backlog of four to seven frames. These are the same deficit wearing different clothes : neither estimator processes frames as fast as they arrive, and the two differ in what they do about it.

46. *table row:* E | replay rate 0.5× for both visual estimators | accuracy figures are not evidence of real-time capability

47. *table row:* E | baselines are offline, visual estimators ran live | timing-induced error is present on one side of the comparison only

48. *table row:* E | only `allsensor` bags carry wheel encoders | the baseline exists for static featureless-floor scenes only; no dynamic condition

49. *table row:* E | $n = 5$ across four worlds, no condition repeated more than twice | no mean, deviation or interval; between-bag differences are not effects

50. *table row:* E | ORB-SLAM3 drops frames at a depth-5 best-effort subscriber; VINS buffers 100 deep | input delivery is not matched between the two; no timing comparison is admissible

51. *table row:* E | ORB-SLAM3 run-to-run spread unmeasured, one run per configuration | loop-closure differences cannot be attributed; repeats required

52. *table row:* E | loop-closure state absent from `run_metadata.csv` | configurations distinguishable only by inspecting `loop_closing.csv`

53. *table row:* E | origin of the wheel yaw-rate offset untested | the drift is predicted but not yet attributable or correctable ([sec:h1-falsified])

54. *table row:* E | accelerometer characterisation unstable across bags | bias 0.019 to 0.068, scale 1.020 to 1.067 for one simulated sensor

Back matter

1. *table row:* [8pt] **Date of Birth** | [to be completed]

2. *table row:* HIGH SCHOOL | [school, year — to be completed]

3. *table row:* **RESEARCH GRANT** | [to be completed, or "—"]